Area and Perimeter — Area/Perimeter Mix-Ups
Answer Key

Grades 2–3

Quick Answers

1. 22	3. 20	5. 28	7. 9
2. 24	4. 18	6. 32	8. —

Common wrong answers

1. *If they got 24: multiplied the two side lengths, which counts the squares inside instead of the distance around*
2. *If they got 22: added all four sides, which measures the distance around instead of the squares inside*
3. *If they got 24: multiplied the length by the width, so the number found was the area and not the perimeter*
4. *If they got 22: added the four sides, so the number found was the perimeter and not the area*
5. *If they got 49: multiplied a side by itself, which fills the garden with squares instead of running fence around it*
6. *If they got 60: multiplied the two sides, which tells how much mat covers the floor rather than how much edging fits around it*
7. *If they got 0: compared the distances around instead of the space inside, and so reported no difference at all*

1. Ribbon around an 8 cm by 3 cm card — find the perimeter.

Step 1: read the question for the deciding word. "All the way around the outside edge" is the distance around, so this is *perimeter*, and perimeter is added, never multiplied.
Step 2: a rectangle has two long sides and two short sides: $8 + 3 + 8 + 3$, which is the same as $2 \times 8 + 2 \times 3 = 16 + 6$.
Step 3: the ribbon needed is $\boxed{22 \text{ cm}}$.
Watch for: a student who multiplies gets 24 — that is the area of the card, not the ribbon around it. The units give it away: ribbon is measured in cm, covering is measured in cm^2.

2. One-centimetre squares covering the same 8 cm by 3 cm card — find the area.

Step 1: "covers the whole card" means we are counting the squares that fit inside, so this is *area.*

Step 2: the squares sit in 3 rows with 8 squares in each row, so the count is 8×3 — multiply, do not add.

Step 3: the card is covered by $\boxed{24 \text{ cm}^2}$.

Watch for: a student who adds the four sides gets 22 — that is the ribbon length from problem 1, not the number of squares. Problems 1 and 2 use the same rectangle on purpose: the two answers are different because the two questions are different.

3. Find and fix: Milo's photo, 6 cm by 4 cm.

Step 1: name what Milo actually computed. 6×4 multiplies the two side lengths, and multiplying length by width gives *area.* So Milo's 24 is the area in cm^2 — a correct number, answering the wrong question.

Step 2: the question asked for perimeter, so add all four sides: $6+4+6+4$, or $2\times 6+2\times 4 = 12 + 8$.

Step 3: the perimeter is $\boxed{20 \text{ cm}}$.

Full-credit answer says: "Milo found the area instead of the perimeter," and then shows the addition.

4. Find and fix: Sara's rug, 9 m by 2 m.

Step 1: name what Sara actually computed. $9 + 2 + 9 + 2 = 22$ adds the four sides, and adding the sides gives *perimeter.* So Sara's 22 is the distance around the rug in m, not the area.

Step 2: the question asked for area, so multiply the two sides: 9×2.

Step 3: the area is $\boxed{18 \text{ m}^2}$.

Full-credit answer says: "Sara found the perimeter instead of the area," and then shows the multiplication. Notice her answer was even labelled "square m" while the work she did produced a plain length — the unit is the fastest check.

5. Fence around a square garden with side 7 m.

Step 1: "a fence goes all the way around" is the perimeter. A square has four equal sides, so instead of $7 + 7 + 7 + 7$ we can use 4×7.

Step 2: $4 \times 7 = 28$.

Step 3: the fence needed is $\boxed{28 \text{ m}}$.

Watch for: $7 \times 7 = 49$. A square is the shape where this mix-up is easiest, because "times" feels natural — but 49 counts the square metres of soil inside the garden, which is not what a fence measures.

6. Edging around a 10 m by 6 m play mat — which one?

Step 1: circle **perimeter**. "Along every outside edge" describes a trip around the border, so the mat's inside is not being counted at all.

Step 2: add around the rectangle: $10 + 6 + 10 + 6$, or $2 \times 10 + 2 \times 6 = 20 + 12$.

Step 3: the edging needed is $\boxed{32 \text{ m}}$.

Watch for: $10 \times 6 = 60$. That is how much mat covers the floor. Edging is bought by the metre, so 60 would send Mr. Ruiz home with almost twice the edging he needs.

7. Ana's 5 m square vs Ben's 8 m by 2 m rectangle: how much more growing space?

Step 1: check the claim in the problem. Ana's fence is $4 \times 5 = 20$ m; Ben's fence is $8 + 2 + 8 + 2 = 20$ m. Equal perimeters, exactly as stated.

Step 2: "growing space" is the space inside, so this question is about *area*, and the areas do not have to match just because the fences do. Ana: $5 \times 5 = 25$ square m. Ben: $8 \times 2 = 16$ square m.

Step 3: subtract the two areas: $25 - 16$, so Ana has $\boxed{9 \text{ m}^2}$ more growing space.

Watch for: a student who subtracts the perimeters gets $20 - 20 = 0$ and answers "no difference." Same perimeter never means same area — that is the whole point of this problem.

8. Show what you know: two rectangles, same perimeter, different areas.

Open task — flagged for manual review. A model answer is below; accept any correct pair and any equivalent sentence.

(a) and **(b)** Model pair, both with perimeter 16 units:

- a 7 by 1 rectangle: perimeter $7 + 1 + 7 + 1 = 16$ units, area $7 \times 1 = 7$ square units.
- a 5 by 3 rectangle: perimeter $5 + 3 + 5 + 3 = 16$ units, area $5 \times 3 = 15$ square units.

Other correct pairs with perimeter 16: 6 by 2 (area 12) and 4 by 4 (area 16). Any two rectangles whose side pairs add to the same half-perimeter work. Full credit needs the perimeter labelled in units and the area labelled in square units on *both* drawings.

(c) Model sentence: "If the question is about going around the outside — fence, ribbon, edging, border — I find the perimeter by adding the sides; if it is about covering the inside — tiles, paint, grass, squares — I find the area by multiplying."

What a full-credit answer must contain: two rectangles that really are different, matching perimeters shown by the student's own addition, two different areas shown by multiplication, correct units on each, and a sentence that names a deciding word rather than just restating the two formulas.